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**MIL-HDBK-1655**
**17 March 1997**
**SUPERSEDING**
**MIL-STD-1655**
**10 April 1975**

# DEPARTMENT OF DEFENSE HANDBOOK

# CLASSIFICATION OF DEFECTS OF FLARELESS AND BEAM SEAL FITTINGS

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AMSC N/A FSC 4730

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# FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. This handbook cannot be cited as a requirement. If it is, the contractor does not have to comply.

3. This standard classifies various defects of flareless and beam seal fittings and defines rejection and acceptance criteria and Quality Assurance Provisions.

4. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-1, Highway 547, Lakehurst, NJ 08733-5100 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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# 1. SCOPE

1.1 Scope. This handbook establishes a classification of commonly occurring defects applicable to flareless fittings and beam seal fittings specified in military specifications MIL-F-18280, MIL-F-85421, MIL-F-85720, and DoD adopted non-Government standards.

1.2 Classification. This handbook defines the classification of defects, critical defects, major defects, and minor defects of parts standards and design standards based upon the effect on usability.

1.3 Purpose. This handbook lists the defects in tabular format, class of defects, applicable defect number, description of defects and references in relation to the corresponding MS drawings to identify the defective parts.

1.4 Applicability. The requirements specified herein apply to all flareless fittings and beam seal fittings specified in military specifications MIL-F-18280, MIL-F-85421, and MIL-F-85720 for use in aerospace fluid systems.

# 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed below are not necessarily all of the documents referenced herein, but are the ones that are needed in order to fully understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

| | | |
|---|---|---|
| MIL-F-18280 | - | Fittings Flareless Tube, Fluid connection |
| MIL-F-85421/1 | - | Fitting End, Standard Dimensions for Beam Seal, Male |
| MIL-F-85421/2 | - | Fitting End, Standard Dimensions for Bulkhead, Beam Seal, Male |
| MIL-F-85421/3 | - | Tee, Reducer, Beam Seal, 3000/4000 psi, 3 Male Ends |
| MIL-F-85421/4 | - | Tee, Reducer, Beam Seal, 3000/4000 psi, Male on Run, Swivel Nut on Branch |
| MIL-F-85421/5 | - | Tee, Reducer, Beam Seal, 3000/4000 psi, Male on Run and Branch, Swivel Nut on Run |
| MIL-F-85421/10 | - | Adapter, Beam Seal, 3000/4000 psi, Male to Flareless |



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| | | |
|---|---|---|
| MIL-F-85421/11 | - | Union, Beam Seal, 3000/4000 psi, Male Ends |
| MIL-F-85421/12 | - | Elbow, 90°, Beam Seal, 3000/4000 psi, Male to Male |
| MIL-F-85421/13 | - | Elbow, 90°, Beam Seal, 3000/4000 psi, Bulkhead, Male Ends |
| MIL-F-85421/14 | - | Elbow, 90°, Beam Seal, 3000/4000 psi, Bulkhead, Male to Female |
| MIL-F-85421/15 | - | Elbow, 45°, Beam Seal, 3000/4000 psi, Bulkhead, Male Ends |
| MIL-F-85421/16 | - | Tee, Beam Seal, 3000/4000 psi, Bulkhead on Run, 3 Male Ends |
| MIL-F-85421/17 | - | Tee, Beam Seal, 3000/4000 psi, Bulkhead on Branch, Male Ends |

STANDARDS

DEPARTMENT OF DEFENSE

| | | |
|---|---|---|
| MS21900 | - | Adapter, Flareless Tube to AN Flared Tube |
| MS21902 | - | Union, Flareless Tube |
| MS21903 | - | Union, Flareless Tube, 3/8 Bulkhead and Universal |
| MS21904 | - | Elbow, Flareless Tube, 90° |
| MS21905 | - | Tee, Flareless Tube |
| MS21906 | - | Cross, Flareless Tube |
| MS21907 | - | Elbow, Flareless Tube and Universal 45° |
| MS21908 | - | Elbow, Bulkhead Universal 90 Deg., Flareless Tube |
| MS21909 | - | Tee, Bulkhead and Universal, Flareless Tube |
| MS21910 | - | Tee, Flareless Tube, Internal Thread on Side |
| MS21911 | - | Tee, Bulkhead Flareless Tube, Internal Thread on Run |
| MS21912 | - | Tee, Flareless Tube with Bulkhead on Run |
| MS21913 | - | Plug, Flareless Tube |
| MS21914 | - | Cap, Pressure Seal, Flareless Tube Fitting |
| MS21915 | - | Bushing, Screw-Thread Expander, Flareless Tube Connection |
| MS21916 | - | Reducer, External Thread, Flareless Tube |
| MS21921 | - | Nut, Sleeve Coupling, Flareless |
| MS21922 | - | Sleeve, Coupling, Flareless |
| MS21923 | - | Adapter, Flareless Tube, Bulkhead and Universal to Flared Tube |
| MS21924 | - | Union, Flareless Tube, Bulkhead and Universal |
| MS21925 | - | Elbow, 90° Universal, Flareless Tube, High Profile |
| MS21926 | - | Elbow, 90° Universal, Flareless Tube, Low Profile |
| MS21933 | - | Fitting End, Bolt Cluster Fitting, Single Port, Through, Flareless |
| MS21935 | - | Fitting End, Bolt Cluster Fitting, Double Port, Through, Flareless |
| MS21937 | - | Nut, Cluster Fitting, Retainer |
| MS21938 | - | Bolt, Cluster Fitting, Single Port, Flareless |



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| | | |
|---|---|---|
| MS21939 | - | Bolt, Cluster Fitting, Single Port, Through, Flareless |
| MS21940 | - | Bolt, Cluster Fitting, Double Port, Flareless |
| MS21941 | - | Bolt, Cluster Fitting, Double Port, Through, Flareless |
| MS21942 | - | Body, Cluster Fitting, One Way, Flareless |
| MS21943 | - | Body, Cluster Fitting, Two Way, 90°, Flareless |
| MS21944 | - | Body, Cluster Fitting, Two Way, 180°, Flareless |
| MS21945 | - | Body, Cluster Fitting, Three Way, Flareless |
| MS24405 | - | Adapter, Flareless Tube to AN Flared Tube, Precision Type |
| MS24651 | - | Elbow, Tube - 90°, Bulkhead, Flareless Tube Universal to Flared Tube |
| MS24652 | - | Elbow, Tube - 90°, Bulkhead, Flareless Tube Universal to Flareless Tube |
| MS24654 | - | Elbow, Tube - 45°, Bulkhead, Flareless Tube Universal to Flared Tube |
| MS33514 | - | Fitting End, Standard Dimensions for Flareless Tube Connection and Gasket Seal |
| MS33515 | - | Fitting End, Standard Dimensions for Bulkhead Flareless Tube Connections |

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein.

PUBLICATIONS

AIR FORCE - NAVY AERONAUTICAL BULLETIN

| | | |
|---|---|---|
| ANA-431 | - | Fittings, Flared, Classification of Defects |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the Contracting Officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the latest issue of the DoDISS and supplement thereto.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

| | | |
|---|---|---|
| ANSI/ASQC Z1.4 | - | Sampling Procedures and Tables for Inspection by Attributes (DoD adopted) |



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(Application for copies should be addressed to the American National Standards Institute (ANSI) 11 West 42nd Street, New York, NY 10036)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AS STANDARDS (SAE)

| | | |
|---|---|---|
| SAE-AS1001 | - | Tee, Reducer, Bulkhead on Run, Internal Port on Side, Flareless Tube (DoD adopted) |
| SAE-AS1002 | - | Tee, Reducer, Bulkhead and Internal Port on Run, Flareless Tube (DoD adopted) |
| SAE-AS1003 | - | Tee, Reducer, Bulkhead on Run, Flareless Tube (DoD adopted) |
| SAE-AS1004 | - | Elbow, 90°, Reducer, Flareless Tube (DoD adopted) |
| SAE-AS1005 | - | Tee, Reducer, Flareless Tube (DoD adopted) |
| SAE-AS1006 | - | Cross, Reducer, Flareless Tube (DoD adopted) |
| SAE-AS1007 | - | Union, Reducer, Flareless Tube (DoD adopted) |
| SAE-AS1008 | - | Elbow, 90°, Reducer, Bulkhead, Flareless Tube (DoD adopted) |
| SAE-AS1009 | - | Tee, Reducer, Bulkhead on Side, Flareless Tube (DoD adopted) |
| SAE-AS1010 | - | Elbow, 45°, Reducer, Flareless Tube (DoD adopted) |

(Applications for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15098.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. DEFINITIONS

3.1 Acceptable Quality Level (AQL). When a continuous series of lots is considered, the AQL is the quality level which, for the purposes of sampling inspection, is the limit of a satisfactory process average (see 4.3.1).

3.2 Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness.

3.3 Critical defect. A critical defect is a defect which reduce materially the usability of the product for its intended purposes and resulted in failure of the system or equipment or both system and equipment.



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3.4 Defect. A defect is any nonconformance of the unit of product with specified requirements.

3.5 Imperfect thread. The runout thread on externally threaded parts is incomplete and less than one pitch in length, causing abrupt change in cross sectional area.

3.6 Inspection. Inspection is the process of measuring, examining, testing, or otherwise comparing the unit of product with the requirements.

3.7 Leg length. The length of the port, male or female, measured from the end of the part to the center of the body vertically or horizontally.

3.8 Major defect. A major defect is a defect, other than critical, that will have effect on usability that is likely to result in failure if used, or it will reduce materially the usability of the unit of product for its intended purpose.

3.9 Minor defect. A minor defect is a defect other than major, essentially no effect on usability, that is not likely to reduce materially the usability of the unit of product for its intended purpose, or having little bearing on the effective use or operation of the unit.

3.10 Not measurement sensitive document. A document in which application of the requirements does not depend substantially on some measured quantity. This type of document can be used with either a metric system or an inch-pound system.

3.11 Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

3.12 Shape fitting. A fitting where the body of fitting is not in a straight line and flow of fluid within the inside diameter of fitting is at an angle. Fittings such as tees, elbows, and crosses are classified as shape fittings.

3.13 Straight fitting. A body of fitting is in a straight line and flow of fluid within inside diameter of fitting is at a straight line direction. Fittings such as adapters, unions, and plugs are classified as straight fittings.

3.14 Thread lead. The entering end of external threads and the entering end of internal threads is outside of the specified limits of size for a length not to exceed two pitches including chamfer.

## 4. GENERAL REQUIREMENTS

4.1 Classification of defects. Defects are herein considered for classification in accordance with the effect they have on safety and usability.



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4.2 Acceptability of classifications. These classifications should govern the acceptance inspection of fittings when so indicated in the fitting specification or drawing or in the acquisition document, and should be applied to inspection of fittings manufactured under such specifications or drawings.

4.3 Sampling. When acceptance inspection for these parts is done by sampling, the sampling plan will be specified in the appropriate acquisition document. If no specific document is specified, the sampling plan will be in accordance with ANSI-Z1.4, and the AQL will be as specified in the contract.

4.3.1 Acceptable quality level (AQL). The acceptable quality level is applied by class of defects. The acceptable quality level should be provided in the contract by the contracting activity or acquiring activity. The following AQL has been used successfully in the past and is recommended for use.

Critical - 0 percent
Major - 1.5 percent
Minor - 4.0 percent

## 5. DETAILED REQUIREMENTS

5.1 Classification of defects.

5.1.1 MS33514 flareless end design standard.

TABLE I. Classification of defects of MS33514.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | X | Distance from gauge dia to tube stop | |
| | 102 | T | Thread per specification | |
| | 103 | - | 24° Seat angle | |
| Major | 201 | - | Surface finish - seat | |
| | 202 | F | Undercut dia | |
| | 203 | - | Surface finish - F dia | |
| | 204 | W | Squareness of face to threads | |
| | 205 | Q | Undercut width | |
| | 206 | - | Concentricity - seat to threads | |
| | 207 | A | Dia - ID | |
| | 208 | D | Dia - tube stop to seat dia | |
| | 209 | - | 75° - tube stop angle | |
| | 210 | E | Distance across hex flats | |



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| | | | | |
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| | 211 | H | Seat depth | |
| | 212 | J | Thread or end length | |
| | 213 | Y | Thread chamfer dia | |
| Minor | 301 | K | Hex chamfer dia | |
| | 302 | M | Rad ID at tube stop | |
| | 303 | V | Rad - Corner at tube stop | |
| | 304 | - | R1 radius - F dia and hex face | R1 = |
| | 305 | - | 15° - Hex chamfer angle | .016/.031 |
| | 306 | - | 45° - Thread chamfer angle | |
| | 307 | - | Imperfect threads | |
| | 308 | - | R2 - radius - thread chamfer at F dia | R2 = |
| | 309 | - | Concentricity - F dia to thread | .005/.015 |

5.1.2 MS33515 Bulkhead flareless end design standard.

TABLE II. Classification of defects of MS33515.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | X | Distance from gauge dia to tube stop | |
| | 102 | T | Thread per specification | |
| | 103 | - | 24° - Seat angle | |
| | 201 | - | Surface finish - seat (micro inches) | 63 $R_a$ max |
| Major | 202 | F | Undercut diameter | - |
| | 203 | - | Surface finish - F dia (micro inches) | 63 $R_a$ min |
| | 204 | W | Squareness of face to threads | |
| | 205 | N | Undercut width | |
| | 206 | - | Concentricity - seat to threads | |
| | 207 | - | Continuation of same thread lead | |
| | 208 | A | Tube - ID | |
| | 209 | D | Tube stop to seat dia | |
| | 210 | - | 75° - Tube stop angle | |
| | 211 | E | Distance across hex flats | |
| | 212 | H | Seat depth | |
| | 213 | K | Distance from shoulder to seat end | |
| | 214 | E | Hex chamfer dia | |
| | 215 | M | Rad - ID at tube stop | |
| | 216 | V | Rad - Corner at tube stop | |
| | 217 | Y | Thread chamfer dia | |
| | 218 | - | Concentricity - F dia to thread | |



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| | 219 | P | Distance from center undercut to seat end | |
| Minor | 301 | - | Radius - G dia and shoulder | |
| | 302 | - | 15° - Hex chamfer angle | |
| | 303 | - | 45° - Thread chamfer angle | |
| | 304 | L | .031 - Shoulder length | |
| | 305 | - | Radius - Thread chamfer at F and G dia | 1/ |
| | 306 | R | Flange width | |
| | 307 | S | Flange dia | |
| | 308 | J | Shoulder dia | |
| | 309 | G | Back undercut dia | |
| | 310 | Q | Back undercut width | |
| | 311 | - | Radius - J dia and flange | 2/ |

1/ Rf = .005/.015, Rg = .015/.030
2/ Rg = .005/.010

5.1.3 MS21921 Nut.

TABLE III. Classification of defects of MS21921.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | T | Thread in accordance with specification | |
| | 102 | E | Small ID | |
| | 103 | - | Concentricity - E and K diameters to thread | |
| | | - | Material per specification or as in contract | |
| Major | 201 | A | Overall length | |
| | 202 | B | OD turn length | |
| | 203 | - | 20° - Hex chamfer angle at B length | |
| | 204 | - | 15° - Hex chamfer angle at tail | |
| | 205 | H | Hex chamfer dia at tail | |
| | 206 | C | OD. turn diameter | |
| | 207 | - | .016 - Corner break width on C diameter | |
| | 208 | - | 45° - Corner break angle on C diameter | |
| | 209 | D | Thread chamfer dia | |
| | 210 | F | Full thread length | |
| | 211 | G | Distance from inside shoulder to front | |
| | 212 | K | Surface finish - internal 45° seat (micro inches) | 125 $R_a$ min |
| | 213 | - | Finish - per specification | |
| Minor | 301 | - | 90° - Thread chamfer angle | |
| | 302 | K | Internal 45° seat dia | |
| | 303 | - | 45° - Internal seat angle | |



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| | 304 | - | Radius - Seat angle and E dia | |
| | 305 | - | .005 - Corner break width at E dia and tail | |
| | 306 | - | 45° - Corner break single at E dia and tail | |
| | 307 | - | Break edges and remove burrs | |
| | 308 | - | Marking - per specification | |

5.1.4 Shape fittings, military standards. MS21904, MS21905, MS21906, MS21907, MS21908, MS21909, MS21910, MS21911, MS21912, MS21925, MS21926, MS24651, MS24652, MS24654.

TABLE IV. Classification of defects of shape fittings, military standards.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | - | Incomplete, incorrect or missing holes | |
| | 102 | - | Internal burrs | |
| | 103 | - | Fitting end per design standard | |
| | 104 | - | Material per specification or as in contract | |
| Major | 201 | - | Leg length, short flareless end | |
| | 202 | - | Leg length, bulkhead flareless end | |
| | 203 | - | Leg length, short flared end | |
| | 204 | - | Leg length, bulkhead flared end | |
| | 205 | - | Leg length, boss end | |
| | 206 | - | Drill depth, short flareless end | |
| | 207 | - | Drill depth, bulkhead flareless end | |
| | 208 | - | Drill depth, short flared end | |
| | 209 | - | Drill depth, bulkhead flared end | |
| | 210 | - | Drill depth, boss end | |
| | 211 | - | Distance across wrench pads | |
| | 212 | - | Angle between legs | |
| | 213 | K | Thread length of high profile flareless end | 1/ |
| | 214 | K | End length of universal low profile flareless end | 2/ |
| | - 215 | - | Finish per specification | |
| Minor | 301 | R | External corner radius forging on elbows | |
| | 302 | U - | Filet corner radius of forging on OD of legs at intersections | |
| | 303 | H | Radius at back of flange on bulkhead end | |
| | 304 | - | Perfect thread depth on boss end | |
| | 305 | Z | OD dia of boss end | |



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| | | | | |
|---|---|---|---|---|
| | 306 | K | Flange thickness at boss end | |
| | 307 | K | Break edges and remove burrs | |
| | 308 | - | Marking per specification | |

1/ Applicable to MS21925 only.
2/ Applicable to MS21926 only.

5.1.5 Shape fittings. DoD adopted non-Government standards AS1001, AS1002, AS1003, AS1004, AS1005, AS1006, AS1008, AS1009, AS1010.

TABLE V. Classification of defects of shape fittings, DoD adopted non-Government standards.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | - | Incomplete, incorrect or missing holes | |
| | 102 | - | Internal burrs | |
| | 103 | - | Fitting end per design std | |
| | 104 | - | Material per specification or as in contract | |
| Major | 201 | - | Leg length, short flareless end | |
| | 202 | - | Leg length, bulkhead flareless end | |
| | 203 | - | Leg length, short flared end | |
| | 204 | - | Leg length, bulkhead flared end | |
| | 205 | - | Leg length, boss end | |
| | 206 | - | Drill depth, short flareless end | |
| | 207 | - | Drill depth, bulkhead flareless end | |
| | 208 | - | Drill depth, short flared end | |
| | 209 | - | Drill depth, bulkhead flared end | |
| | 210 | - | Drill depth, boss end | |
| | 211 | - | Distance across wrench pads | |
| | 212 | - | Angle between legs | |
| | 213 | - | Finish per specification | |
| Minor | 301 | - | Perfect thread depth on boss end | |
| | 302 | - | OD dia of boss end | |
| | 303 | - | Flange thickness at boss end | |
| | 304 | - | Break edges and remove burrs | |
| | 305 | - | Marking per specification | |

5.1.6 Straight fittings, military standards. MS21900, MS21902, MS21903, MS21913, MS21915, MS21916, MS21923, MS21924, MS24405.


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TABLE VI. Classification of defects of straight fittings, military standards.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | - | Incomplete, incorrect or missing holes | |
| | 102 | - | Internal burrs | |
| | 103 | - | Fitting end per design standards | |
| | 104 | - | Material per specification or as in contract | |
| | 105 | T | Thread per specification | |
| Major | 201 | - | Overall length | |
| | 202 | - | Length, end to hex face, bulkhead flareless end | |
| | 203 | - | Shoulder width, bulkhead flareless end | W= .313 |
| | 204 | Q | Undercut width, flareless end dia | 1/ |
| | 205 | - | Drill depth, short flareless end | |
| | 206 | - | Drill depth, bulkhead flareless end | |
| | 207 | - | Drill depth, internal port | |
| | 208 | - | Distance across hex flats | |
| | 209 | - | Surface finish, undercut dia and hex face, | |
| | - | - | flareless end | 1/ |
| | 210 | - | Finish per specification | |
| Minor | 301 | - | 15° - Hex chamfer angle | 2/ |
| | 302 | - | Groove depth in hex corners | 1/ |
| | 303 | - | Groove position in hex corners | 1/ |
| | 304 | - | Groove angle in hex corners | 1/ |
| | 305 | - | Break edges and remove burrs | |
| | 306 | - | Marking per specification | |

1/ Applicable to MS24405 only
2/ Applicable to MS21913 and MS21915

5.1.7 Straight fittings, DoD adopted non-Government standard. AS1007

TABLE VII. Classification of defects of straight fittings DoD adopted non-Government standard.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | - | Incomplete, incorrect or missing holes | |
| | 102 | - | Internal burrs | |
| | 103 | - | Fitting end per design standard | |



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| | | | | |
|---|---|---|---|---|
| | 104 | - | Material per specification or as in contract | |
| | 105 | T | Thread T in accordance with specification | |
| Major | 201 | A | Overall length | |
| | 202 | - | Drill depth, short flareless end | |
| | 203 | - | Drill depth, bulkhead flareless end | |
| | 204 | - | Distance across hex flats | |
| | 205 | - | Surface finish, undercut dia and hex face | |
| | 206 | - | Finish per specification | |
| Minor | 301 | - | Break edges and remove burrs | |
| | 302 | - | Marking per specification | |

5.1.8 MS21914 cap assembly.

TABLE VIII. Classification of defects of MS21914.

| Class of defect | Defect No | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | - | Fitting end per design standard | |
| | 102 | T | Thread per specification | |
| | 103 | - | Material per specification or as in contract | |
| Major | 201 | - | Nut free to swivel | |
| | 202 | - | Finish per specification | |
| | 203 | W | Retainer wire within hex limits | |
| | 204 | L | Length of assembly | |
| | 205 | B | Hex nut | |
| | 206 | C | Diameter | |
| | 207 | A | Length of insert | |
| Minor | 301 | - | Break edges and remove burrs | . |

5.1.9 MS21933 Single port cluster bolt end design standard.

TABLE IX. Classification of defects of MS21933.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | K | Sealing lands dia | |
| | 102 | - | Surface finish - sealing lands | |
| | - | - | (micro inch) | 16 $R_a$ |



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| | | | | |
|---|---|---|---|---|
| | 103 | | Fitting end per design standard | |
| Major | 201 | A | End length | |
| | 202 | B | Length over lands | |
| | 203 | C | Length to front land | |
| | 204 | D | Cross hole position | |
| | 205 | E | Length to back land | |
| | 206 | F | Land chamfer length | |
| | 207 | G | Cross hole dia | |
| | 208 | H | Front undercut dia | |
| | 209 | J | Center relief dia | |
| | 210 | L | Back undercut dia | |
| | 211 | - | Surface finish per specification | |
| | 212 | - | Concentricity - J and K dia to thread | .005 TIR |
| Minor | 301 | - | 45° - Back thread chamfer | |
| | 302 | - | .135/.090 - Front undercut width | |
| | 303 | - | .056/.021 Rad - Front undercut radii | |
| | 304 | - | .094/.031 Rad - Land chamfer radii | |
| | 305 | - | 30° - Land chamfers | |
| | 306 | - | .026 Rad - Back undercut radius | |
| | 307 | - | .052 - Back undercut width | |

5.1.10 MS21935 Double port cluster bolt end design standard.

TABLE X. Classification of defects of MS21935.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | M | Sealing lands dia | |
| | 102 | - | Surface finish - sealing lands(micro inch) | 16 $R_a$ min |
| | 103 | - | Fitting end per design standard | |
| Major | 201 | A | End length | |
| | 202 | B | Length over lands | |
| | 203 | C | Length to front land | |
| | 204 | D | Length to front cross hole | |
| | 205 | E | Length to front of center land | |
| | 206 | F | Length to back of center land | |
| | 207 | G | Length to back cross hole | |
| | 208 | H | Length to back land | |
| | 209 | J | Front undercut dia | |
| | 210 | K | Cross holes dia | |
| | 211 | L | Relief dia | |
| | 212 | N | Back undercut dia | |
| | 213 | P | Land chamfer length | |



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| | | | | |
|---|---|---|---|---|
| | 214 | - | Surface finish per specification | .005 TIR |
| | 215 | - | Concentricity - L and M dia to thread | |
| Minor | 310 | - | 45° - Back thread chamfer | |
| | 311 | - | .090/.135 - Front undercut width | |
| | 312 | - | .056/.021 Rad - Front undercut radii | |
| | 313 | - | .094/.031 Rad - Land chamfer radii | |
| | 314 | - | 30° - Land chamfers | |
| | 315 | - | .026 Rad - Back undercut radius | |
| | 316 | - | .052 - Back undercut width | |

5.1.11 MS21922 Sleeve.

TABLE XI. Classification of defects of MS21922.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | - | Surface finish, shoulder, internal dia. and sharp | |
| | - | - | cutting edge | |
| | 102 | - | 35° - shoulder angle | |
| | 103 | H | OD of shoulder | |
| | 104 | - | Material per specification or as in contract | |
| Major | 201 | B | Distance from shoulder back to pilot end | |
| | 202 | D | OD of tail | |
| | 203 | E | ID | |
| | 204 | G | OD of barrel | |
| | 205 | - | .020 Rad - Front of shoulder and barrel | |
| | 206 | C | Width of shoulder | |
| | 207 | - | Concentricity - D.E.F. and G dia | |
| | 208 | A | Overall length | |
| | 209 | - | .030 - Groove width | |
| | 210 | - | .030 - Groove location from shoulder | |
| | 211 | - | .010/.005 Groove depth | |
| | 212 | - | Finish - per specification | |
| Minor | 301 | J | Dia - ID. of pilot after spinning | |
| | 302 | - | Radius - ID. and OD at tail | |
| | 303 | - | Radius - Shoulder angle and tail | R at 2° angle |
| | 304 | - | 12° - Pilot angle after spinning (see note) | |
| | 305 | - | Marking - per specification | |

Note: The following items are not inspectable after spinning the pilot to 12° after machining:



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F Dia - ID of pilot before spinning
K - Distance from edge to pilot end
R Rad - Edge and pilot ID
S Rad - Barrel OD at pilot end
2° - Cutting edge angle

5.1.12 MS21937 Nut.

TABLE XII. Classification of defects of MS21937.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | T | Thread per specification | |
| | 102 | | Material per specification or as in contract | |
| Major | 201 | A | Overall length | |
| | 202 | B | Hex chamfer dia | |
| | 203 | C | Countersink dia | |
| | 204 | D | Angle of lock-wired hole | |
| | 205 | E | Counter bore dia | |
| | 206 | F | Outside turn dia | |
| | 207 | H | Distance across hex flats | |
| | 208 | - | Finish - per specification | |
| Minor | 301 | - | .156 - length of outside turn dia | |
| | 302 | - | 15° - angle of hex chamfer | |
| | 303 | - | Corner of hex chamfer and turn dia | .031 Rad |
| | 304 | | 45° - angle of countersink | |
| | 305 | - | .165/.125 - length of counter bore | |
| | 306 | - | .062 Dia - lock-wired hole dia | |
| | 307 | - | .031 - Position of lock-wired hole | |
| | 308 | - | Break edges and remove burrs | |
| | 309 | - | Marking - per specification | |

5.1.13 Cluster bolts, single and double port. MS21938, MS21939, MS21940, MS21941.



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TABLE XIII. Classification of defects of MS21938, MS21939, MS21940, MS21941.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | - | Incomplete, incorrect or missing holes | |
| | 102 | - | Internal burrs | |
| | 103 | - | Fitting end per design standard | |
| | 104 | - | Material per specification or as in contract | |
| | 105 | T | Thread per specification | |
| Major | 201 | - | Overall length | |
| | 202 | - | Drill depth, past cross hole | |
| | 203 | - | Drill depth, short flareless end | |
| | 204 | - | Drill depth, long flareless end | |
| | 205 | - | Drill hole dia | |
| | 206 | - | Surface finish, outside machined surfaces | |
| | 207 | - | Surface finish, inside surfaces | |
| Minor | 301 | - | Hex chamfer angle | |
| | 302 | - | Hex chamfer dia | |
| | 303 | - | Break edges and remove burrs | |
| | 304 | - | Marking per specification | |
| | 305 | - | Finish per specification | |

5.1.14 Cluster bodies, one, two, and three way. MS21942, MS21943, MS21944, MS21945.

TABLE XIV. Classification of defects of MS21942, MS21943, MS21944, MS21945.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | - | Incomplete, incorrect or missing holes | |
| | 102 | - | Internal burrs | |
| | 103 | - | Fitting end per design standard | |
| | 104 | - | Material per specification or as in contract | |
| | 105 | T | Thread per specification | |
| Major | 201 | E | Leg length | |
| | 201 | - | Angle between legs | 90° ±2 ½° |
| | 202 | C | Leg turn length | |
| | 203 | - | Surface finish machined surfaces | |



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| | | | | |
|---|---|---|---|---|
| | 204 | - | Finish per specification | |
| Minor | 301 | - | Chamfer angle at body and leg | 10° ± 5° |
| | 302 | - | .031 - Radius at body and leg | |
| | 303 | D | Chamfer dia at body and leg | |
| | 304 | - | Break edges and remove burrs | |
| | 305 | - | Marking per specification | |

5.1.15 Beam seal end design standards. MIL-F-85421/1, MIL-F-85421/2, MIL-F-85720/1(AS)

TABLE XV. Classification of defects of MIL-F-85421/1, MIL-F-85421/2, MIL-F-85720/1(AS).

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | - | Sealing surface per specification | |
| | 102 | F | Angle F° | |
| | 103 | T | Thread per specification | |
| | 104 | B | Outer rim sealing surface | |
| Major | 201 | - | .015/.000 undercut at thread relief | |
| | 202 | - | .030/.015 radius undercut | |
| | 203 | - | .031 dimension bulkhead | |
| | 204 | J | Dimension | 1/ |
| | 205 | - | Concentricity- A, B, C to threads | .005 FIM |
| | 206 | C | Diameter | |
| | 207 | - | Finish per specification | |
| | 208 | A | Tube ID | |
| | 209 | E | Dimension thread length | |
| | 210 | H | Dimension | |
| Minor | 301 | G | Dimension | |
| | 302 | - | 45° - chamfer dia | |
| | 303 | M | Hex chamfer dia | |
| | 304 | D | Dimension | |
| | 305 | - | Break edges and remove burrs | |
| | 306 | - | Marking per specification | |

1/ Applicable to MIL-F-85421/2 only

5.1.16 Straight fittings, beam seal, military standards. MIL-F-85421/10,



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MIL-F-85421/11

TABLE XVI. Classification of defects of straight fittings, beam seal, military standards.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | - | Fitting end per design standard | |
| | 102 | T | Thread per specification | |
| | 103 | - | Material per specification or as in contract | |
| Major | 201 | G | Dimension | |
| | 202 | B | Dimension | |
| | 203 | - | Machined surface per specification | |
| | 204 | - | Concentricity D, H, and J to thread | |
| | 205 | - | Surface finish per specification | |
| | 206 | E | Hex - across flats | |
| | 207 | - | Overall length | |
| Minor | 301 | - | 45° ± 5° chamfer | |
| | 302 | F | Diameter | |
| | 303 | C | Hex width | |
| | 304 | - | 15° ± 5° chamfer | |
| | 305 | - | Marking per specification | |

5.1.17 Shape fittings, beam seal, military standards. MIL-F-58421/3, MIL-F-8421/4, MIL-F-85421/5, MIL-F-85421/12, MIL-F-85421/13, MIL-F-85421/14, MIL-F-85421/15, MIL-F-85421/16, MIL-F-85421/17.

TABLE XVII. Classification of defects of shape fittings, beam seal, military standards.

| Class of defect | Defect No. | References on drawing | Items to be examined | Notes |
|---|---|---|---|---|
| Critical | 101 | - | Fitting ends per design standard | |
| | 102 | - | Distance, sealing surface to center body | |
| | 103 | T | Thread per specification | |
| | 104 | | Material per specification or as in contract | |
| Major | 201 | - | Leg length port 1 | |
| | 202 | - | Leg length port 2 | |
| | 203 | - | Leg length port 3 | |
| | 204 | - | Angle between legs | |
| | 205 | - | Surface finish per specification | |



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| | | | | |
|---|---|---|---|---|
| | 206 | - | Hex dimension across flats | |
| Minor | 301<br>302<br>303 | E<br>-<br>- | Radius<br>.030/.125 Break radius<br>Marking -per specification | |

5.2 Boss fitting end (MS33649) and flared fitting end design standards. Classification of defects for boss fitting end and flared fitting end design standards will be found in ANA 431 bulletin.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory).

6.1 Intended use. This handbook contains inspection and test guidelines for beam seal and flareless fittings acquired by Government Agencies that are intended for use in aerospace fluid systems. Documents conforming to the requirements of this handbook are intended for use as DoD standardization documents and are listed in the DoDISS. This handbook established a common format throughout the DoD for the preparation of flareless fitting and beam seal fitting specifications and standards citing classification of defects.

6.2 Issue of DoDISS. When this handbook is used in acquisition, the applicable issue of the DoDISS should be cited in the solicitation (see 2.1 and 2.2.1).

6.3 Superseding data. This handbook supersedes MIL-STD-1655 dated 10 April 1975.

6.4 Subject term (keyword) listing.

Critical defect
Defect
Defective
Examination
Fluid connector
Inspection
Major defect
Minor defect
Quality assurance

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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Air Force - 99

Review activities:
Navy - MC
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Preparing activity:
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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-HDBK-1655 | 2. DOCUMENT DATE (*YYMMDD*)<br>1997 March 17 |
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3. DOCUMENT TITLE
CLASSIFICATION OF DEFECTS OF FLARELESS AND BEAM SEAL FITTINGS

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

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